HEART DISEASE PREDITION

Project Report

Submitted by

ABIN N M

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CHAPTER 1

INTRODUCTION

Heart disease is one of the leading causes of mortality worldwide. Early detection plays a critical role in reducing fatal outcomes and improving patient survival rates. With the advancement of data analytics and machine learning, predictive systems can assist healthcare professionals in diagnosing heart disease more accurately and efficiently.

The dataset used in this project consists of 1025 patient records with 14 clinical attributes such as age, cholesterol level, blood pressure, chest pain type, and maximum heart rate achieved. The target variable indicates whether a patient has heart disease (1) or not (0).

The significance of this dataset lies in its ability to help build predictive models that can identify patterns in clinical measurements and determine the likelihood of heart disease. This project applies statistical analysis, visualization techniques, and machine learning algorithms to develop a predictive system.

CHAPTER 2

DATA PREPROCESSING

Data preprocessing is a crucial step in machine learning to ensure data quality and improve model performance.

2.1 HANDLING MISSING VALUES

The dataset was checked for missing values using the `isnull()` function. No missing values were found in any column, ensuring data completeness.

2.2 DUPLICATE REMOVAL

Duplicate records were removed using:

```
df.drop_duplicates(inplace=True)
```

This step prevents biased learning caused by repeated entries.

2.3 OUTLIER REMOVAL USING IQR

Outliers were detected and removed using the Interquartile Range (IQR) method.

$$IQR = Q3 - Q1$$

Lower Bound:

$$Q1 - 1.5 \times IQR$$

Upper Bound:

$$Q3 + 1.5 \times IQR$$

Any data point outside this range was removed.

After applying outlier removal, dataset size reduced from:

$$1025 \rightarrow 228$$

This indicates significant elimination of extreme values.

2.4 FEATURE SELECTION

The target variable was separated from independent features:

```
X = df.drop('target', axis=1)
y = df['target']
```

2.5 FEATURE SCALING

StandardScaler was used to normalize feature values:

$$Z = \frac{X - \mu}{\sigma}$$

Scaling ensures that features with larger magnitudes do not dominate the model.

CHAPTER 3

EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand patterns, relationships, and distributions in the dataset.

3.1 TARGET DISTRIBUTION

A count plot was used to visualize the distribution of heart disease cases.

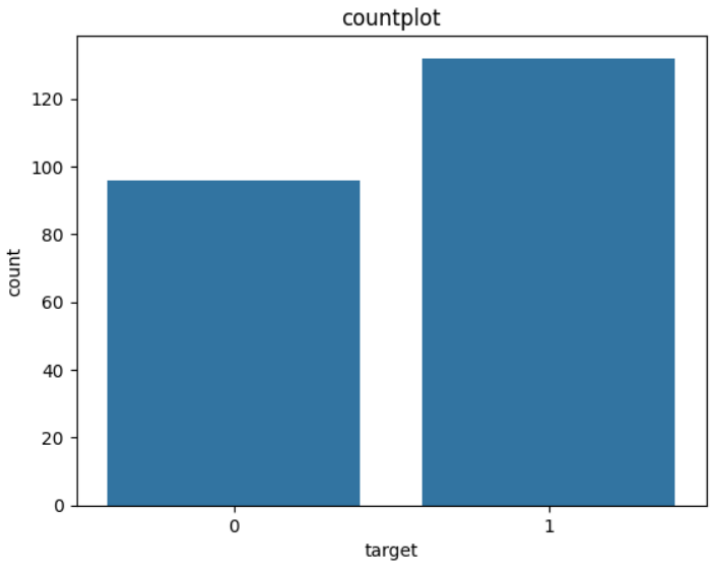

Figure 3.1: *Target Distribution*

The visualization shows the number of patients with and without heart disease, helping determine class balance.

3.2 CORRELATION HEATMAP

A correlation heatmap was plotted to analyze relationships between variables.

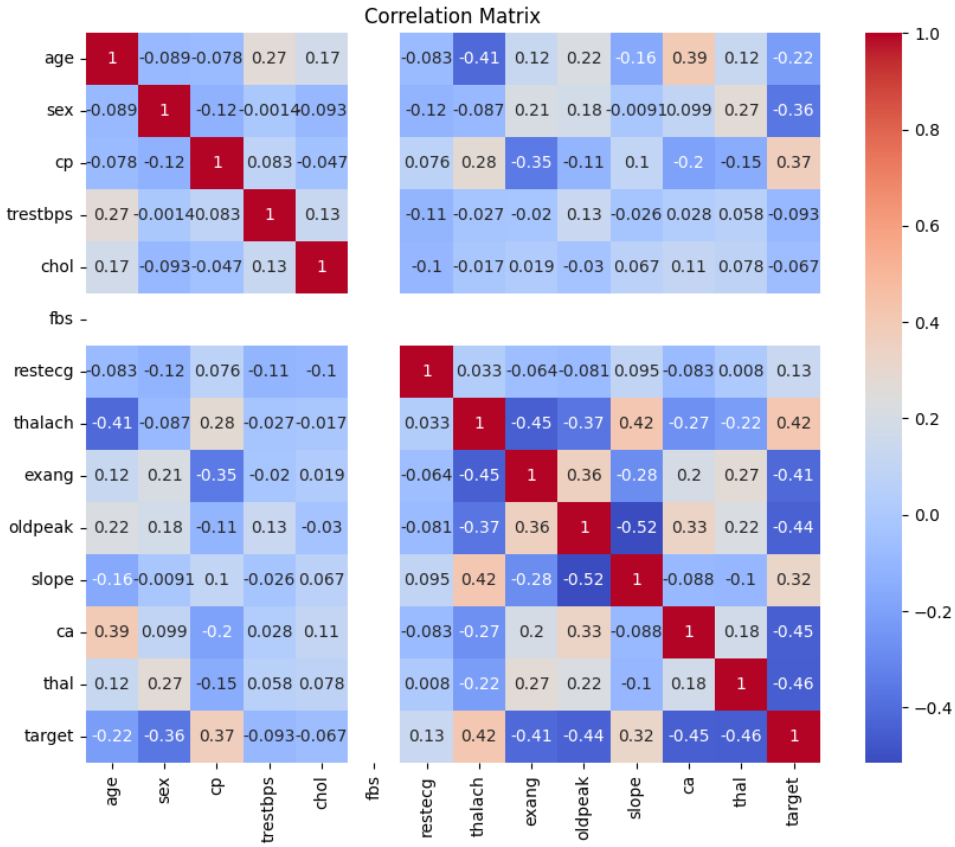

Figure 3.2: *Correlation Heatmap*

Strong correlations were observed between:

- thalach and target
- oldpeak and target
- chest pain type and target

3.3 BOXPLOT VISUALIZATION

Boxplots were generated to identify spread and detect outliers in numerical features.

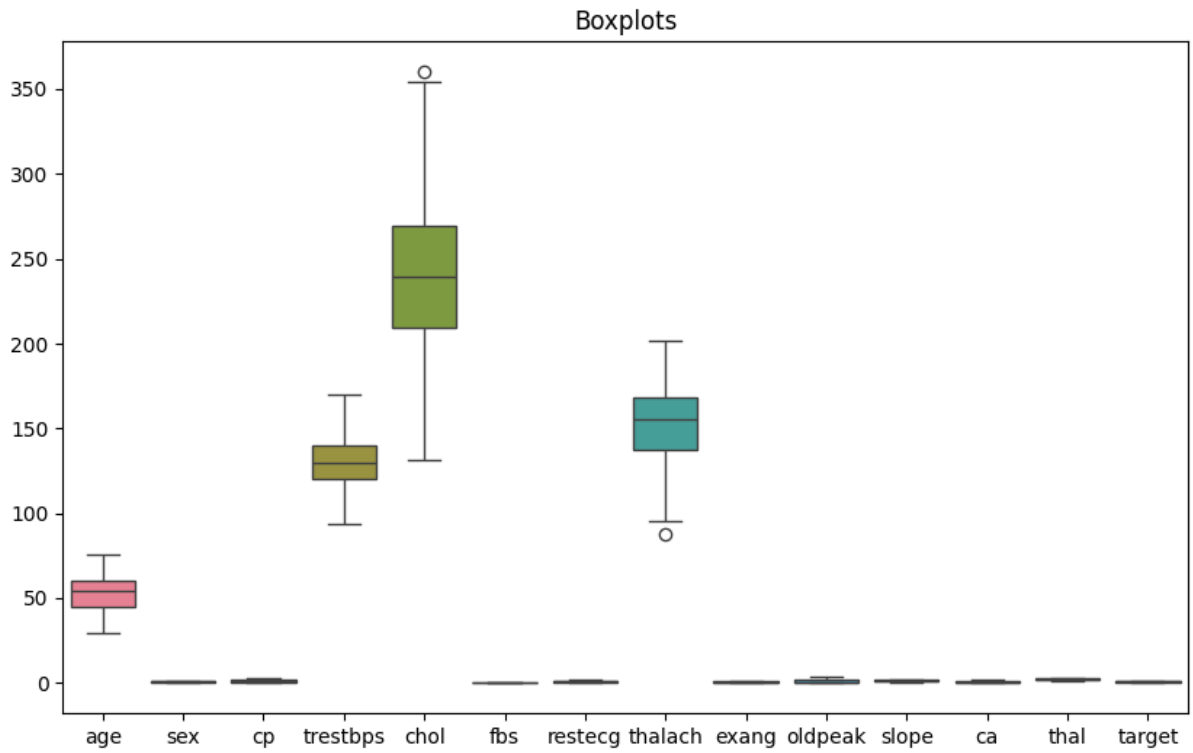

Figure 3.3: *Boxplot of Features*

3.4 HISTOGRAM ANALYSIS

A histogram was used to analyze age distribution.

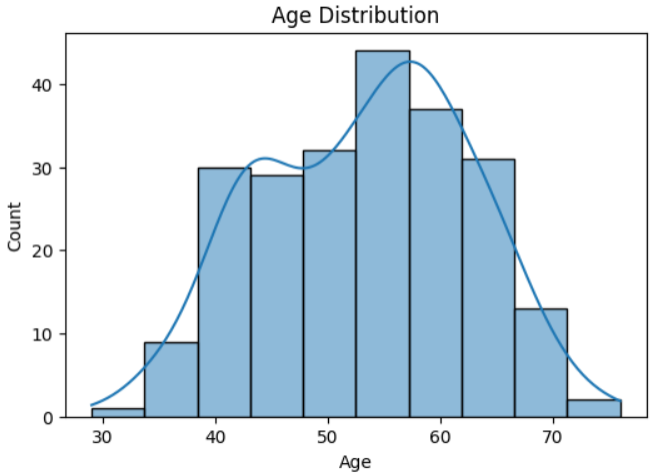

Figure 3.4: *Age Distribution*

Most patients fall within the 40–60 age range.

3.5 SCATTER PLOT (AGE VS CHOLESTEROL)

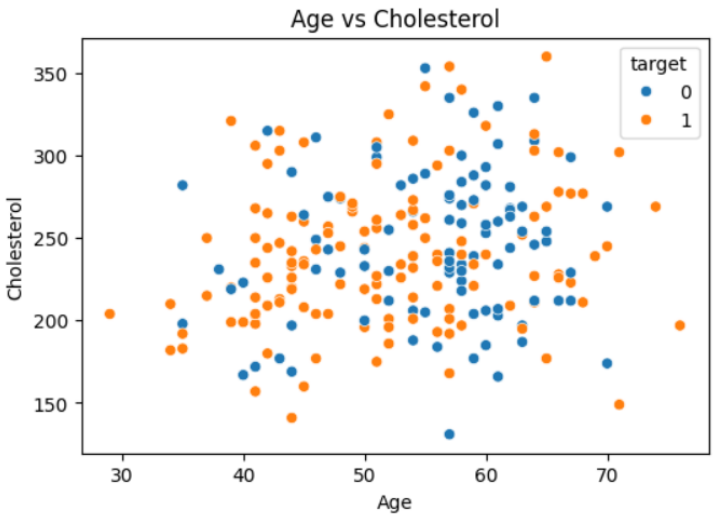

Figure 3.5: *Age vs Cholesterol*

The scatter plot shows clustering behavior between diseased and non-diseased patients.

3.6 PAIR PLOT ANALYSIS

A pair plot was generated to visualize pairwise relationships between selected numerical features and the target variable.

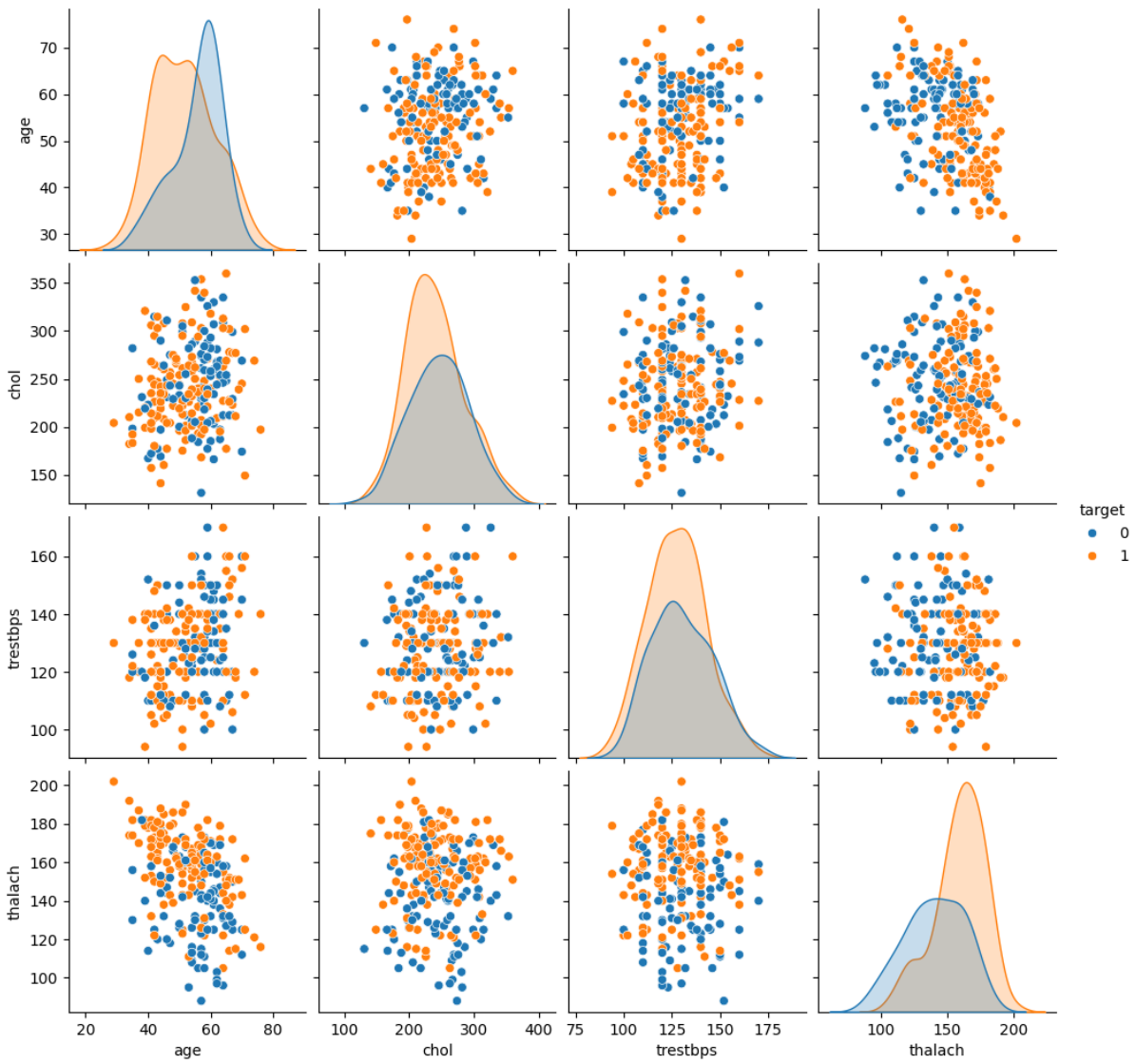

Figure 3.6: *Pair Plot of Selected Features*

The pair plot provides the following insights:

- It visualizes pairwise relationships between features such as age, cholesterol, resting blood pressure, and maximum heart rate.

- The diagonal plots show the distribution of each feature.
- The scatter plots highlight class separation when colored by the target variable.
- Certain features such as thalach (maximum heart rate) show noticeable separation between patients with and without heart disease.

This visualization helps in identifying feature interactions, trends, and potential patterns useful for model training.

CHAPTER 4

MODEL SELECTION & TRAINING

The dataset was divided into 80% training data and 20% testing data using train-test split.

4.1 LOGISTIC REGRESSION

Logistic Regression is a binary classification algorithm that uses the sigmoid function:

$$h(x) = \frac{1}{1 + e^{-z}}$$

Where:

$$z = wX + b$$

It outputs probability values between 0 and 1.

Reasons for selection:

- Suitable for binary classification
- Simple and interpretable
- Efficient for small datasets

4.2 RANDOM FOREST

Random Forest is an ensemble algorithm that builds multiple decision trees and aggregates their predictions.

```
Accuracy of model : 0.8695652173913043

Details of confusion matrix :
              precision    recall  f1-score   support

           0       0.80      0.80      0.80        15
           1       0.90      0.90      0.90        31

    accuracy                           0.87        46
   macro avg       0.85      0.85      0.85        46
weighted avg       0.87      0.87      0.87        46
```

Figure 4.1: *Random Forest*

Reasons for selection:

- Handles non-linear relationships
- Reduces overfitting
- High robustness and accuracy

Both models were trained using scaled training data.

CHAPTER 5

EVALUATION & RESULTS

5.1 ACCURACY

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Logistic Regression Accuracy: 84.78%

Random Forest Accuracy: 84.78%

5.2 CLASSIFICATION METRICS

```
Accuracy of model : 0.8695652173913043

Details of confusion matrix :
              precision    recall  f1-score   support

           0       0.80      0.80      0.80        15
           1       0.90      0.90      0.90        31

    accuracy                           0.87        46
   macro avg       0.85      0.85      0.85        46
weighted avg       0.87      0.87      0.87        46
```

Figure 5.1: *Classification Report (Random Forest)*

5.3 CONFUSION MATRIX

The confusion matrix shows:

- True Positives
- True Negatives
- False Positives

- False Negatives

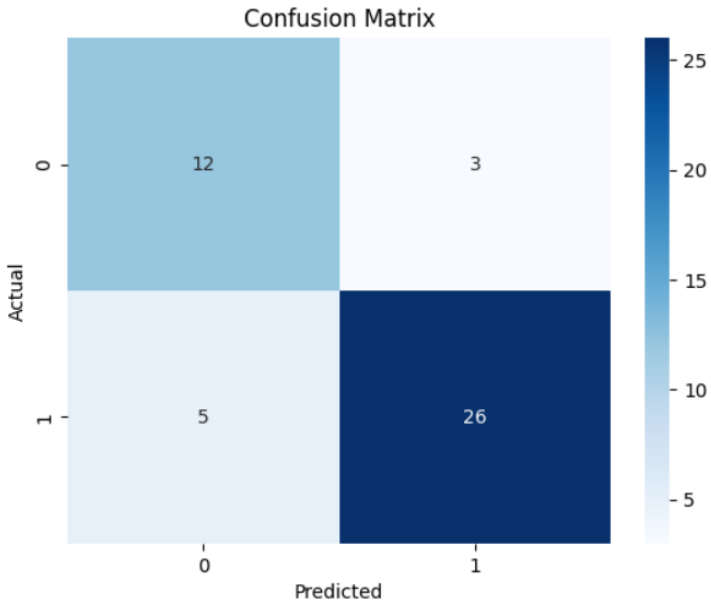

Figure 5.2: *Confusion Matrix*

5.4 ROC CURVE

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

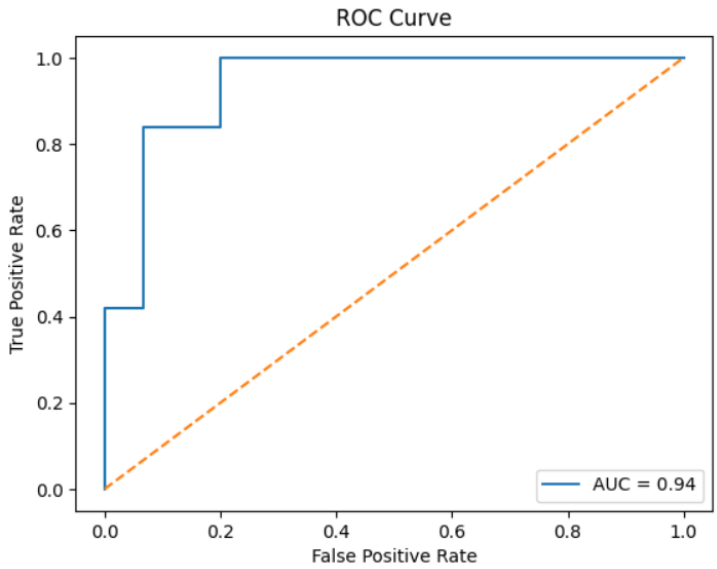

Figure 5.3: *ROC Curve*

The ROC curve indicates good classification ability.

CHAPTER 6

CONCLUSION

This project successfully analyzed heart disease data and built predictive models using Logistic Regression and Random Forest.

Key findings:

- Data preprocessing significantly improved data quality.
- Outlier removal reduced dataset size substantially.
- Both models achieved 84.78% accuracy.
- Random Forest showed strong precision in detecting heart disease.

Machine learning techniques can assist medical professionals in early detection and risk assessment of heart disease.

CHAPTER 7

PROJECT EVIDENCE

7.1 IMPLEMENTATION SCREENSHOTS

```python
#Importing neccessary Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, roc_curve, auc

#Dataset loaded
df = pd.read_csv("heart.csv")
print(df.head())
print("\nDataset Info:")
print(df.info())
print("\nNull values:")
print(df.isnull().sum())
print("\n")
#Duplicates handled
df.drop_duplicates(inplace=True)
#Outlier handled
def remove_outliers(df):
    numeric_cols = df.select_dtypes(include=np.number).columns
    for col in numeric_cols:
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR
        df = df[(df[col] >= lower) & (df[col] <= upper)]
    return df
df = remove_outliers(df)
print("Shape without outlier:", df.shape)
print("\n")

#Target distribution
sns.countplot(x='target', data=df)
plt.title("countplot")
plt.show()
print("\n")
```

Figure 7.1: *Project Implementation Screenshot 1*

```python
# correlation matrix calculation
corr_matrix = df.corr()
print("Correlation Matrix:\n")
print(corr_matrix)
print("\n")
# Visualize correlation matrix
plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix, cmap='coolwarm', annot=True)
plt.title("Correlation Matrix")
plt.show()
print("\n")

# Boxplots
plt.figure(figsize=(10,6))
sns.boxplot(data=df, orient="v")
plt.title("Boxplots")
plt.show()
print("\n")

# Histogram
plt.figure(figsize=(6,4))
sns.histplot(data=df, x='age', kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
print("\n")
# Scatter plot
plt.figure(figsize=(6,4))
sns.scatterplot(data=df, x='age', y='chol', hue='target')
plt.title("Age vs Cholesterol")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.show()
print("\n")

#Feature and Target Split
X = df.drop('target', axis=1)
y = df['target']
print("\n")
#Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
print("\n")
```

Figure 7.2: *Project Implementation Screenshot 2*

```python
#Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
print("\n")

#Model Training
lr_model = LogisticRegression(max_iter=1000)
lr_model.fit(X_train, y_train)
print("\n")

# Random Forest
rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)
print("\n")
#Predictions
lr_pred = lr_model.predict(X_test)
rf_pred = rf_model.predict(X_test)
print("\n")
#Model Evaluation
print("\nAccuracy of model :",accuracy_score(y_test, rf_pred))
print("\nDetails of confusion matrix :")
print(classification_report(y_test, rf_pred))
print("\n")
#Confusion Matrix
cm = confusion_matrix(y_test, rf_pred)
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
print("\n")
#ROC Curve
rf_probs = rf_model.predict_proba(X_test)[:,1]
fpr, tpr, _ = roc_curve(y_test, rf_probs)
roc_auc = auc(fpr, tpr)
plt.plot(fpr, tpr, label="AUC = %0.2f" % roc_auc)
plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
print("\n")
#pair plot
selected_features = ["age", "chol", "trestbps", "thalach", "target"]
sns.pairplot(df[selected_features], hue="target")
plt.show()
```

Figure 7.3: *Project Implementation Screenshot 3*

7.2 OUTPUT SCREENSHOTS

```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   52    1   0       125   212    0        1      168      0      1.0      2
1   53    1   0       140   203    1        0      155      1      3.1      0
2   70    1   0       145   174    0        1      125      1      2.6      0
3   61    1   0       148   203    0        1      161      0      0.0      2
4   62    0   0       138   294    1        1      106      0      1.9      1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1025 non-null   int64
 1   sex       1025 non-null   int64
 2   cp        1025 non-null   int64
 3   trestbps  1025 non-null   int64
 4   chol      1025 non-null   int64
 5   fbs       1025 non-null   int64
 6   restecg   1025 non-null   int64
 7   thalach   1025 non-null   int64
 8   exang     1025 non-null   int64
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64
 11  ca        1025 non-null   int64
 12  thal      1025 non-null   int64
 13  target    1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
None

Null values:
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64


Shape without outlier: (228, 14)
```


Figure 7.4: *Project Output*

```
X_train shape: (182, 13)
X_test shape: (46, 13)
y_train shape: (182,)
y_test shape: (46,)
```

Figure 7.5: *Test and Train Data*

7.3 PROJECT SOURCE CODE

The complete project source code and related files are available at the following Google Drive link:

https://drive.google.com/drive/folders/1iGrqu0vYBf6BQSonnRZRXvZQpE0enwJc